

① Multiplicative Inverse mod 26

$$\rightarrow 1 = 1^{-1} \text{ mod } 26 = 1$$

$$\rightarrow 3^{-1} \text{ mod } 26 = 9$$

$$\rightarrow 5^{-1} \text{ mod } 26 = 21$$

$$\rightarrow 7^{-1} \text{ mod } 26 = 15$$

$$\rightarrow 9^{-1} \text{ mod } 26 = 3$$

$$\rightarrow 11^{-1} \text{ mod } 26 = 19$$

$$\rightarrow 15^{-1} \text{ mod } 26 = 7$$

$$\rightarrow 17^{-1} \text{ mod } 26 = 23$$

$$\rightarrow 19^{-1} \text{ mod } 26 = 11$$

$$\rightarrow 21^{-1} \text{ mod } 26 = 5$$

$$\rightarrow 23^{-1} \text{ mod } 26 = 17$$

$$\rightarrow 25^{-1} \text{ mod } 26 = 25$$

② Affine Encrypt

"CRYPT"
2 17 24 15 19

$$K_1 = 5$$

$$K_2 = 8$$

5 & 26 are coprime integers.

$$C = (K_1 * P + K_2) \text{ mod } 26$$

$$C = (5 * 2 + 8) \text{ mod } 26 = 18 \text{ mod } 26 = 18 \rightarrow S$$

$$R = (5 * 17 + 8) \text{ mod } 26 = 93 \text{ mod } 26 = 15 \rightarrow P$$

$$Y = (5 * 24 + 8) \text{ mod } 26 = 128 \text{ mod } 26 = 24 \rightarrow V$$

$$P = (5 * 15 + 8) \text{ mod } 26 = 83 \text{ mod } 26 = 5 \rightarrow F$$

$$T = (5 * 19 + 8) \text{ mod } 26 = 103 \text{ mod } 26 = 25 \rightarrow Z$$

SPVFZ

③ Encrypt "HI" using Affine with $a=11$ & $b=5$.
 $11 \& 26$ are coprime

$$H = (11 \cdot 7 + 5) \bmod 26 = 82 \bmod 26 = 4$$

$$I = (11 \cdot 8 + 5) \bmod 26 = 93 \bmod 26 = 15$$

$$C = EP$$

④ ciphertext $^{20}2^{17}$ was encrypted using Affine
 Find plaintext
 $f_{unl} \quad 9x+2 \bmod 26$
 $K_1 \quad K_2$

$$K_1 = 9, K_2 = 2$$

$$D = K_1^{-1}(C - K_2) \bmod 26$$

$$\therefore V = 3(20 - 2) \bmod 26$$

$$9^{-1} \bmod 26 = 3$$

$$= 54 \bmod 26 = 2 = C$$

$$C = 3(2 - 2) \bmod 26 = 0 = A$$

$$R = 3(17 - 2) \bmod 26 = 19 = T$$

$$\therefore P = CAT$$

⑥ Use $K_1 = 13$ & $K_2 = 4$ string $\rightarrow$ "input"
"altu"

13 & 26 are not coprime

* INPUT
8 13 15 -

$$I = (13 \times 8 + 4) \bmod 26 = 108 \bmod 26 = 4 = E$$

$$N = (13 \times 13 + 4) \bmod 26 = 17 \bmod 26 = 17 = R$$

$$P = (13 \times 15 + 4) \bmod 26 = 199 \bmod 26 = 17 = R$$

$$V = (13 \times 20 + 4) \bmod 26 = 264 \bmod 26 = 4 = E$$

$$T = (13 \times 19 + 4) \bmod 26 = 17 = R$$

$$\begin{array}{r} 26 \overline{) 199} \quad 7 \\ 182 \\ \hline 17 \end{array}$$

$\therefore$ ERROR

Since 13 & 26 are not coprime,
many distinct plaintext maps to same
cipher text letter.

④ Inverse matrix for 16y.

$$\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \bmod 26$$

$$1^{st} \quad D = (7 \times 3) - (5 \times 2) = 21 - 10 = 11$$

2nd $11^{-1} \bmod 26 = \underline{\underline{19}}$

3rd Adjacent $A = \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \bmod 26$

$$\begin{array}{r} 26 \overline{) -5} 1 \\ \underline{26} \\ 21 \end{array}$$

$$= \begin{bmatrix} 7 & 24 \\ 21 & 3 \end{bmatrix}$$

$$\begin{array}{r} -2 \bmod 26 \\ 26 \overline{) -2} 1 \\ \underline{26} \\ 24 \end{array}$$

$$A^{-1} = 19 \times \begin{bmatrix} 7 & 24 \\ 21 & 3 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 133 & 456 \\ 399 & 57 \end{bmatrix} \bmod 26 = \boxed{\begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}}$$

⑧ Inverse of

$$A = \begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix}$$

* $D = (11 \times 3) - 26 = 7$

* $7^{-1} \bmod 26 = 15$

* $Adj(A) = \begin{bmatrix} 3 & -13 \\ -2 & 11 \end{bmatrix} \bmod 26 = \begin{bmatrix} 3 & 13 \\ 24 & 11 \end{bmatrix}$

xy by rows

$$A^{-1} = 15 \times \begin{bmatrix} 3 & 13 \\ 24 & 11 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 19 & 17 \\ 2 & 17 \end{bmatrix}$$

⑨ Is the key valid $\begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix}$

key is valid if the Determinant is coprime to 26.

① $D = (42 - 182) = -140$.

② $-140 \pmod{26} = 16$.

③ check $\gcd(16, 26)$.

Since $\gcd(16, 26) = 2 \neq 1$

$\therefore$ not a valid key.

$$\begin{array}{r} 26 \overline{) -140(6} \\ \underline{156} \\ 16 \end{array}$$

⑩ Encrypt Hill cipher

$$\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix}$$

plaintext = 'SOLVED'

$$*_{S_0} \begin{bmatrix} 18 \\ 14 \end{bmatrix} \Rightarrow \begin{bmatrix} 3 \\ 7 \end{bmatrix} \begin{bmatrix} 8 \\ 11 \end{bmatrix} \begin{bmatrix} 18 \\ 14 \end{bmatrix} = \begin{bmatrix} 166 \\ 280 \end{bmatrix} \pmod{26} = \begin{bmatrix} 4 \\ 20 \end{bmatrix}$$

$$*_{L_V} \begin{bmatrix} 11 \\ 21 \end{bmatrix} \Rightarrow \begin{bmatrix} 3 \\ 7 \end{bmatrix} \begin{bmatrix} 8 \\ 11 \end{bmatrix} \begin{bmatrix} 11 \\ 21 \end{bmatrix} = \begin{bmatrix} 201 \\ 308 \end{bmatrix} \pmod{26} = \begin{bmatrix} 19 \\ 22 \end{bmatrix}^T w$$

$$*_{ED} \begin{bmatrix} 4 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \end{bmatrix} \begin{bmatrix} 8 \\ 11 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} = \begin{bmatrix} 36 \\ 61 \end{bmatrix} \pmod{26} = \begin{bmatrix} 10 \\ 9 \end{bmatrix}^T k$$

$$\therefore \boxed{a^0 p_{wskt} = K V T w k J}$$